

Key
44

[0]	[1]	[2]	[3]	[4]	[5]
55	44	22	66	11	33

j i

- 1) Check if j is at valid index ?
 $J \geq 0$? Yes
 &&
 2) $44 < arr[j]$? Yes
 a) Shift $arr[j]$ one step ahead
 b) $j--$

[0]	[1]	[2]	[3]	[4]	[5]
	55	22	66	11	33

j i

Check if j is at valid index ? NO
 Copy key to j+1 index

[0]	[1]	[2]	[3]	[4]	[5]
44	55	22	66	11	33

Key
22

[0]	[1]	[2]	[3]	[4]	[5]
44	55	22	66	11	33

j i

- Check if j is at valid index ?
 $J \geq 0$? Yes
 &&
 $Key < arr[j]$? Yes
 1) Shift $arr[j]$ one index ahead
 2) $j--$

[0]	[1]	[2]	[3]	[4]	[5]
44		55	66	11	33

j i

- $J \geq 0$? Yes
 &&
 $Key < arr[j]$? Yes
 1) Shift $arr[j]$ one index ahead
 2) $j--$

[0]	[1]	[2]	[3]	[4]	[5]
	44	55	66	11	33

j i

$J \geq 0$? NO
 Copy key back to the array
 At j+1 index

[0]	[1]	[2]	[3]	[4]	[5]
22	44	55	66	11	33

Key
66

[0]	[1]	[2]	[3]	[4]	[5]
22	44	55	66	11	33

j i

- $J \geq 0$? Yes
 &&
 $Key < arr[j]$?
 $66 < 55$? NO

Copy key back to the
array
At j+1 index

Key
11

[0]	[1]	[2]	[3]	[4]	[5]
22	44	55	66	11	33

j i

J >= 0 ? Yes
&&
Key < arr[j] ? Yes
1) Shift arr[j] one index
ahead
2) j--

[0]	[1]	[2]	[3]	[4]	[5]
22	44	55		66	33

j i

J >= 0 ? Yes
&&
Key < arr[j] ? Yes
1) Shift arr[j] one index ahead
2) j--

[0]	[1]	[2]	[3]	[4]	[5]
22	44		55	66	33

j i

J >= 0 ? Yes
&&
Key < arr[j] ? Yes
1) Shift arr[j] one index
ahead
2) j--

[0]	[1]	[2]	[3]	[4]	[5]
22		44	55	66	33

j i

j >= 0 &&
Key < arr[j]
1) Shift
2) j--

[0]	[1]	[2]	[3]	[4]	[5]
	22	44	55	66	33

j i

J is not at valid index.
Copy key back to the
array
At j+1 index

[0]	[1]	[2]	[3]	[4]	[5]
11	22	44	55	66	33

Key
33

[0]	[1]	[2]	[3]	[4]	[5]
11	22	44	55	66	33

j i

J >= 0 yes
&& key < arr[j] ? Yes
1) Shift
2) Decrement j

[0]	[1]	[2]	[3]	[4]	[5]
11	22	44	55		66

j i

J >= 0
&&
33 < 55 ? Yes
1) Shift

2) j--

[0]	[1]	[2]	[3]	[4]	[5]
11	22	44		55	66

j

i

J >= 0

&&

33 < 44 ? Yes

1) Shift

2) j--

[0]	[1]	[2]	[3]	[4]	[5]
11	22		44	55	66

j

i

j >= 0 ? Yes

&&

33 < 22 ? NO

[0]	[1]	[2]	[3]	[4]	[5]
11	22	33	44	55	66

j

Copy key back to the
array

At j+1 index